Q18: Project Suggestions

DLS 2026 — Kukla

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1. **Brain-score mini (medium, 4–6 days).** Take a pretrained vision model and a public neural-response dataset; measure how well layer activations linearly predict neural firing. Relate predictivity to layer depth.

2. **Biologically-plausible learning (medium, 1 week).** Implement feedback-alignment (random backward weights) vs. backprop on a small task; quantify the accuracy gap and discuss what it implies for the brain-plausibility debate.

3. **Energy accounting (small, 2–3 days).** Estimate joules-per-inference for a small transformer and compare to the $\sim$20 W human brain budget for a loosely matched task; write up the orders-of-magnitude gap and its sources.